

Skills Summary

Two Different Questions: a Rate and a Starting Value

Use this reference while completing your worksheet or when studying.

Skill 1 — Read the roles, not the order

Slope-intercept form: $y = mx + b$, where m multiplies x and b stands alone.

- m = how much y changes per 1 of x (a rate). b = the value of y when $x = 0$ (one output).
- Order does not decide roles: $12 - 2x$ has $m = -2$ and $b = 12$, exactly like $-2x + 12$.
- **The $x = 0$ test:** substitute $x = 0$. Whatever is left is b .

Example: For $y = 6x + 2$, find y when $x = 3$.

Step 1. Identify the roles before substituting: the number attached to x is the slope, so it is the one that gets multiplied.

Step 2. $y = 6(3) + 2 = 18 + 2$

The value is **20**.

Try it: For $y = 9 - 3x$, find y when $x = 4$.

check

Skill 2 — Slope from two points: y on top

Slope formula: $m = \frac{\text{change in } y}{\text{change in } x}$, and subtract the two coordinates in the *same* order top and bottom.

- Read it as “ y per x ” — the units tell you which goes on top.
- An inverted slope is the reciprocal of the right one, so if your answer times the correct one is 1, you divided the wrong way round.
- A y coordinate is never a slope, and it is the intercept only when its point has $x = 0$.

Example: Find the slope through $(1, 4)$ and $(5, 16)$.

Step 1. Slope is y per x , so put the change in y on top and keep both subtractions in the same order.

Step 2. $m = \frac{16 - 4}{5 - 1} = \frac{12}{4}$

The slope is **3**.

Try it: Find the slope through $(3, 10)$ and $(7, 2)$.

check

Skill 3 — The intercept a table does not show

Backing up to $x = 0$: if the table starts at $x = 1$, then $b = (\text{the } y \text{ at } x = 1) - (\text{the step size})$.

- Find the step size first: subtract consecutive y values (when the x values go up by 1).
- The first row of a table is only the intercept if that row is $x = 0$.
- Check by rebuilding $y = mx + b$ and testing it on a row you were given.

Example: A table has $y = 11$ at $x = 1$ and goes up by 5 for each step of 1 in x . Find the y -intercept.

Step 1. The intercept lives at $x = 0$, one step to the left of the first row, so step backwards by the step size instead of reading the row itself.

Step 2. $b = 11 - 5$

The y -intercept is **6**.

Try it: A table has $y = 20$ at $x = 1$ and goes up by 7 for each step of 1 in x . Find the y -intercept.

check: 13

(!) Watch out: If an equation you wrote fails the very point it came from, you almost certainly used a y coordinate as the intercept. Substitute the point back in — it takes one line and catches the swap every time.
